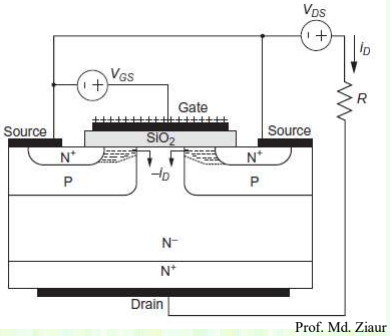
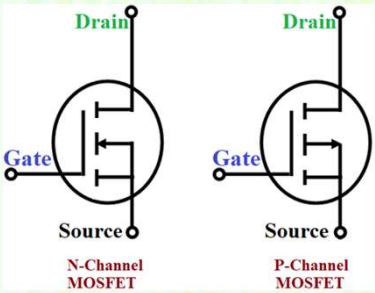
Lecture 6

Power Electronic Devices- Power MOSFET

Power MOSFET is a type of MOSFET which is specially meant to handle high levels of power. These exhibit high switching speed and can work much better in comparison with other normal MOSFETs in the case of low voltage levels. However its operating principle is similar to that of any other general MOSFET. Power MOSFETs which are most widely used are n-channel Enhancement-mode or p-channel Enhancement-mode or n-channel Depletion-mode in nature.

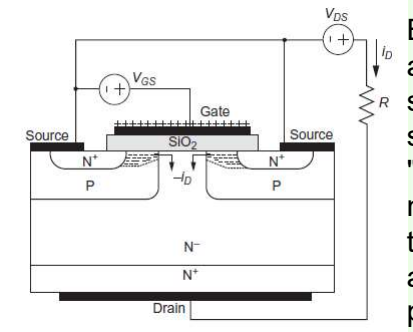




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Power MOSFET

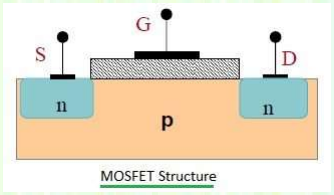
Although early power MOSFET manufacturing used several different designs, the Vertical Diffused MOS (VDMOS) — also known as the Double-diffused MOS (DMOS) — emerged as the predominate design. The terms "vertical" and "diffusion" are adequate to describe the construction of a power MOSFET. "Vertical" refers to the fact that, unlike traditional (or "lateral") MOSFETs, a power MOSFET's source terminal is above its drain terminal, resulting in a primarily vertical current.



By contrast, a lateral MOSFET's source and drain terminals are on opposite sides of the device, with the gate situated on the n-doped channel. "Diffusion" refers to the device's manufacturing process: the p-type and n-type wells that sit on the n-type substrate are created using a double-diffusion process.

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Power MOSFET



Power MOSFET operation is slightly different than that of a lateral e-MOSFET. In lateral MOSFETs, planar construction results in current and breakdown voltage values which are functions of the channel dimensions, rendering the device relatively inefficient. In contrast, the vertical construction of power MOSFETs results in voltage ratings which are functions of the doping and thickness of the n-type well, while current rating remains a function of channel width. Power MOSFETs are therefore able to sustain high voltages and high currents without increasing the size of the device.

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Power MOSFET

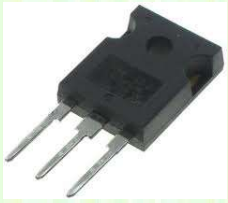



Features of Power MOSFET

- Low input current.
- Voltage controlled device.
- High switching speed.
- No second breakdown problem
- Has high input impedance
- It is a unipolar device.
- For commutation, an extra circuit is not necessary
- Small size
- Needs small power to hold it within the activated condition.

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Power MOSFET

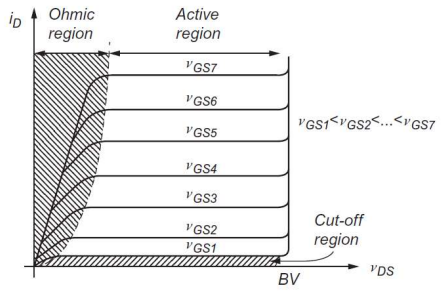



Disadvantage of Power MOSFET

- The on state voltage across a power MOSFET is high. Therefore the on state power dissipation is very high, as compared to that of a power transistor.
- Power MOSFETs need special care while handling otherwise they can get damaged due to static electricity.
- Power MOSFETs have asymmetric blocking capacity. They can block a high forward voltage but they cannot block high reverse voltage

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Power MOSFET



Ohmic Region

In the ohmic region (triode region) or constant resistance region, the channel of the MOSFET behaves like a constant resistance. Such resistance (R_{DSon}) is linearly proportional to drain-source voltage and the drain current.

$$v_{DS} > v_{GS} - V_{TH} \quad and \quad v_{GS} > V_{TH}$$

Cut-off Region

The cutoff region is characterized by the absence of the channel formation.

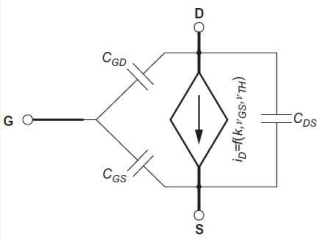
$$v_{DS} > 0 \quad and \quad v_{GS} < V_{TH}$$

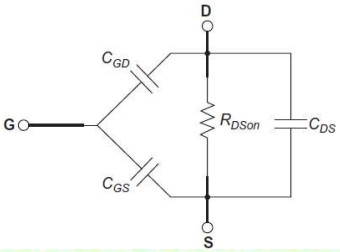
Active Region

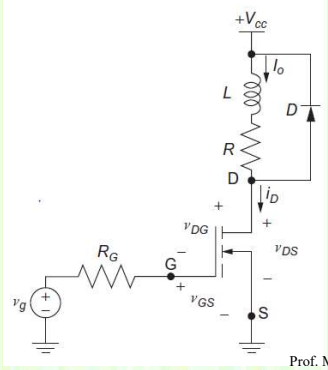
The active region is characterized by a constant drain current, controlled by the gate-source voltage $v_{DS} > v_{GS} - V_{TH} \quad and \quad v_{GS} > V_{TH}$

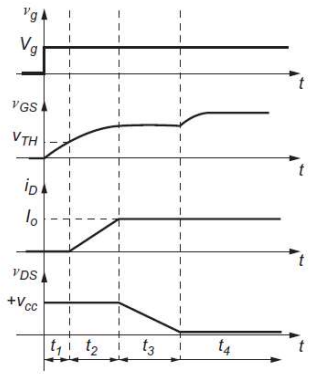
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Power MOSFET - Turn-on Analysis



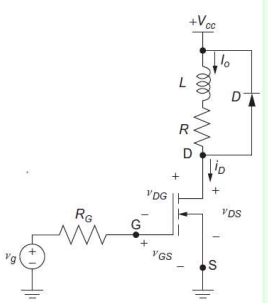






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Power MOSFET - Turn-on Analysis



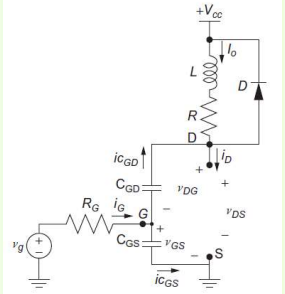


FIG. 2.47 Equivalent circuit for the interval t_1.

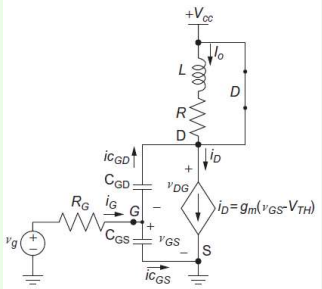
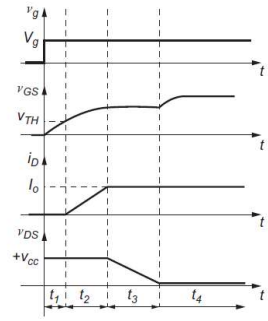


FIG. 2.48 Equivalent circuit for the interval t_2.



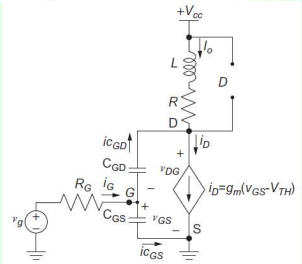


FIG. 2.49 Equivalent circuit for the interval t_3.

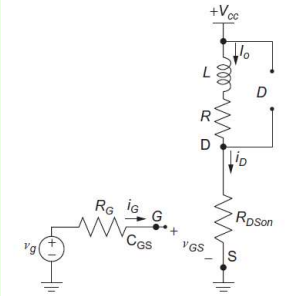


FIG. 2.50 Equivalent circuit for the interval t_4.

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Power MOSFET - Turn-off Analysis

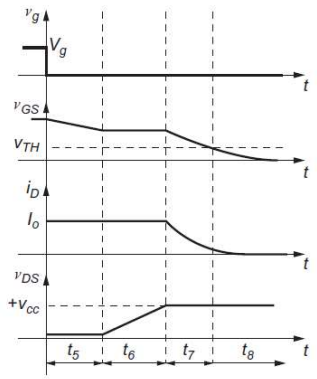


FIG. 2.51 Waveforms during the turn-off process.

During interval t_5, the gate-source voltage begins to fall, discharging the gate-source capacitance and charging the gate-drain capacitance. The drain current and the drain-source voltage keep unchanged.

During interval t_6, the drain-source voltage rises, while the gate-source voltage keeps constant. At the end of interval t_6, the drain-source voltage reaches the value of the DC source ($+V_{cc}$), turning on the diode, and the drain current begins to fall.

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Power MOSFET - Turn-off Analysis

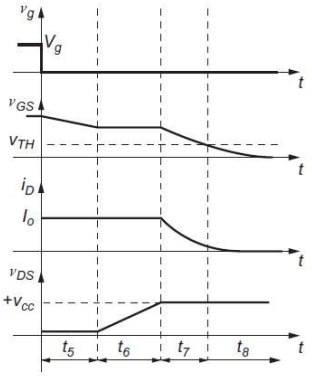


FIG. 2.51 Waveforms during the turn-off process.

The interval t_7, corresponds to the time required to the gate-source voltage to reach the threshold voltage. After that, the drain current is null, and the MOSFET is in the cutoff region.

Similarly to the turn-on behavior, the intervals t_6 and t_7 present considerable drain current and drain-source voltage, contributing to the switching losses.

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Power MOSFET - Safe Operation Area

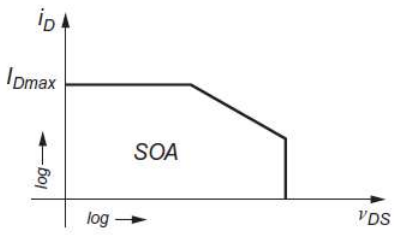
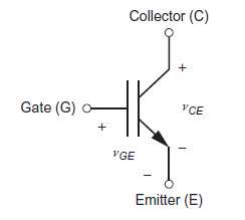


FIG. 2.52 Safe operation area of the MOSFET.

The SOA of the MOSFET is shown in Fig. 2.52. The area SOA shows the values of drain current, and drain-source voltage can hold without destructive failure. MOSFETs do not present the second breakdown.

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Power Electronic Devices: Insulated Gate Bipolar Transistor

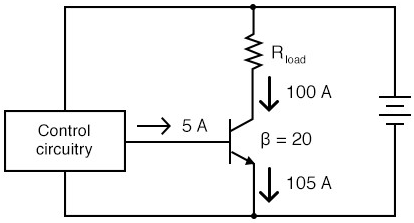


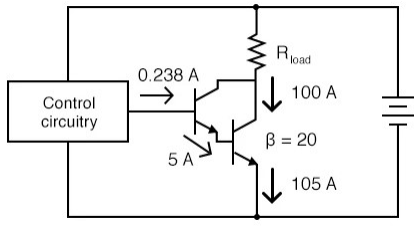


Insulated Gate Bipolar Transistor (IGBT) is a three-terminal semiconductor switching device that can be used for fast switching with high efficiency in many types of electronic devices. These devices are mostly used in amplifiers for switching/processing complex wave patters with pulse width modulation (PWM).

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Insulated Gate Bipolar Transistor



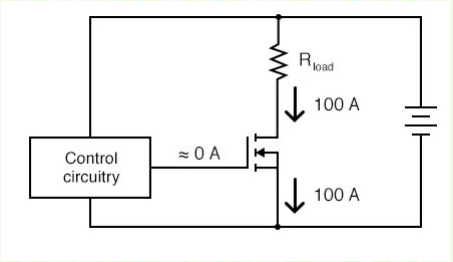


If a bipolar junction transistor is used to control a large collector current, there must be a substantial base current sourced or sunk by some control circuitry

we can use Darlington pair transistors to increase the current gain, but this kind of arrangement still requires far more controlling current than an equivalent power MOSFET.

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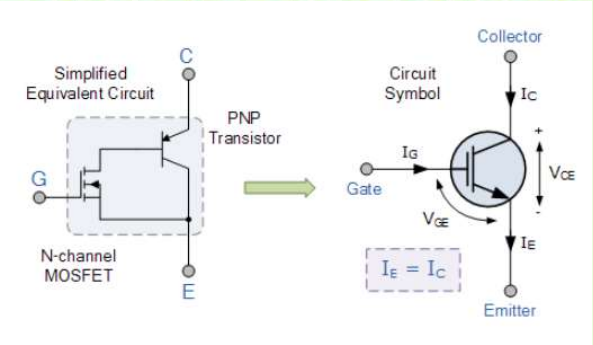
Insulated Gate Bipolar Transistor



Unfortunately, though, MOSFETs have problems of their own controlling high current: they typically exhibit greater drain-to-source voltage drop while saturated than the collector-to-emitter voltage drop of a saturated BJT. This greater voltage drop equates to higher power dissipation for the same amount of load current, limiting the usefulness of MOSFETs as high-power devices. Although some specialized designs such as the so-called VMOS transistor have been designed to minimize this inherent disadvantage, the bipolar junction transistor is still superior in its ability to switch high currents.

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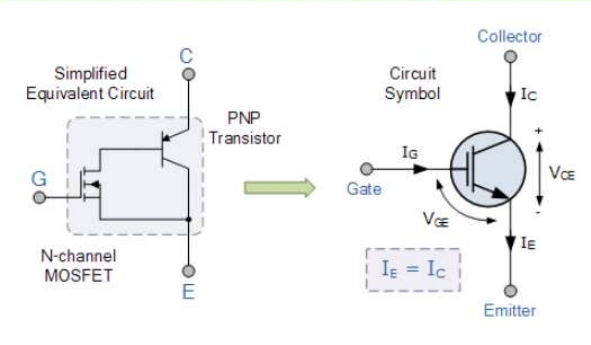
Insulated Gate Bipolar Transistor



An interesting solution to this dilemma leverages the best features of MOSFETs with the best of features of BJTs, in one device called an Insulated-Gate Bipolar Transistor, or IGBT. Also known as an Bipolar-mode MOSFET, a Conductivity-Modulated Field-Effect Transistor (COMFET), or simply as an Insulated-Gate Transistor (IGT), it is equivalent to a Darlington pair of IGFET and BJT:

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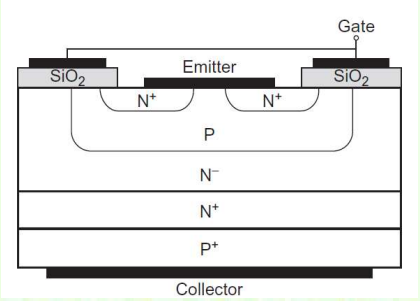
Insulated Gate Bipolar Transistor - Disadvantage

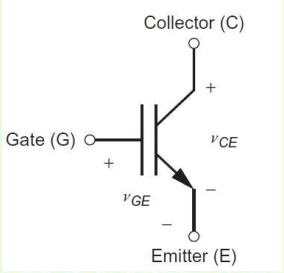


One disadvantage of the IGBT over a standard BJT is its slower turn-off time. For fast switching and high current-handling capacity, it's difficult to beat the bipolar junction transistor. Faster turn-off times for the IGBT may be achieved by certain changes in design, but only at the expense of a higher saturated voltage drop between collector and emitter. However, the IGBT provides a good alternative to IGFETs and BJTs for high-power control applications.

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IGBT

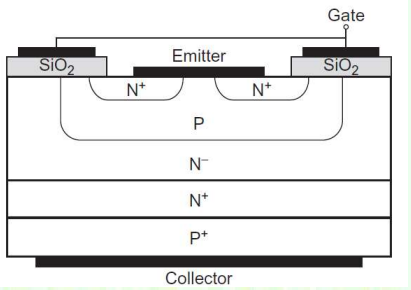


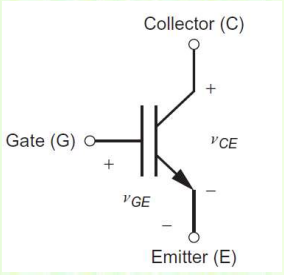


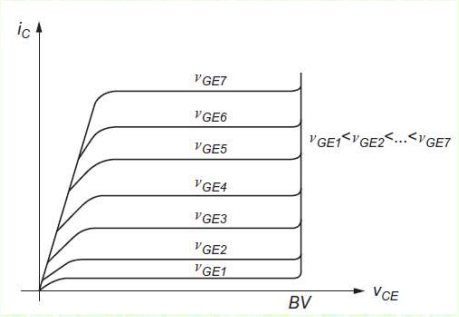
The IGBT has three terminal called collector (C), emitter (E), and gate (G). The gate terminal is isolated from the rest of the semiconductor by means of a SiO_2 layer. The P+, N+N-, and P junctions constitute a parasitic BJT. By the application of a positive voltage higher than the threshold between the gate and emitter, the N-channel is created, similar to MOSFET, providing a path for a current from the collector to the emitter terminal.

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IGBT





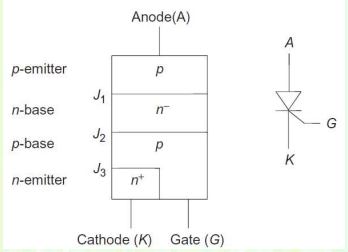


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Next Class

Silicon controlled rectifier(SCR)/Thyristors

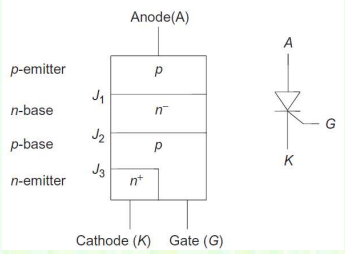
Thyristors are usually three-terminal devices that have four layers of alternating p-type and n-type material (i.e., three p-n junctions) comprising its main power handling section



The control terminal of the thyristor, called the gate (G) electrode, may be connected to an integrated and complex structure as a part of the device. The other two terminals, called the anode (A) and cathode (K), handle the large applied potentials (often of both polarities) and conduct the major current through the thyristor. The anode and cathode terminals are connected in series with the load to which power is to be controlled.

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SCR-Basic Structure and Operation



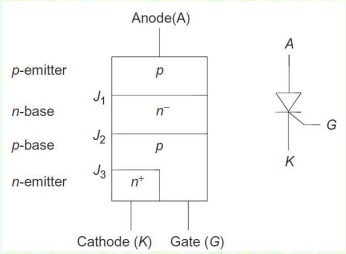
A high-resistivity region, n-base, is present in all thyristors. This region is associated junction, J_2, which must support the large applied forward voltages that occur when the switch is in its off- or forward-blocking state. The n-base is typically doped with impurity phosphorous atoms at a concentration of 10^{13}–10^{14} cm^{-3}.

Doping profile of the p-regions ranges from about 10^{15} to 10^{17} cm^{-3}. These p-regions can be up to tens of micrometer thick. The cathode region (typically only a few micrometers thick) is formed by using phosphorous atoms at a doping density of 10^{17}–10^{18} cm^{-3}.

Operation of thyristors is as follows. When a positive voltage is applied to the anode (with respect to cathode), the thyristor is in its forward-blocking state. The center junction J_2 (see Fig. 3.1) is reverse biased.

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SCR-Basic Structure and Operation

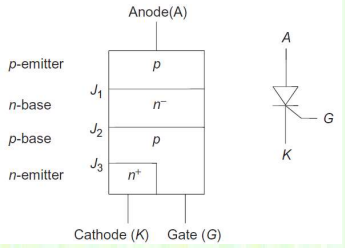


In this operating mode, the gate current is held to zero. In practice, the gate electrode is biased to a small negative voltage to reverse bias the GK junction J_3 and prevent charge carriers from being injected into the p-base. In this condition, only thermally generated leakage current flows through the device and can often be approximated as zero in value.

As long as the forward applied voltage does not exceed the value necessary to cause excessive carrier multiplication in the depletion region around J_2 (avalanche breakdown), the thyristor remains in an off-state (forward-blocking). If the applied voltage exceeds the maximum forward-blocking voltage of the thyristor, it will switch to its on-state. However, this mode of turn-on causes nonuniformity in the current flow, is generally destructive, and should be avoided.

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SCR-Basic Structure and Operation



When a positive gate current is injected into the device, J_3 becomes forward biased and electrons are injected from the n-emitter into the p-base. Some of these electrons diffuse across the p-base and get collected in the n-base. This collected charge causes a change in the bias condition of J_1.

The change in bias of J_1 causes holes to be injected from the p-emitter into the n-base. These holes diffuse across the n-base and are collected in the p-base. The addition of these collected holes in the p-base acts the same as gate current. The entire process is regenerative and will cause the increase in charge carriers until J_2 also becomes forward biased and the thyristor is latched in its on-state (forward conduction).

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